that the other ports are, so that when the piston is at the upper end of the compression stroke the valve is opened and the first gas enters the crank case in the same manner as in position A, Figure 2. Aside from this the operations are exactly the same as in the two-port cycle.

These engines may be reversed by changing the time of ignition. Other advantages are simplicity, light weight, and cheapness. Some of its disadvantages are that it never has a clean mixture: always some of the burned gases are mixed in with the fresh charge. But in well designed engines the amount of this mixture is not serious. The explosions being more frequent, the cooling and lubricating problems are necessarily harder.

For general farm use the four-cycle is to be recommended as being more reliable and more economical in the long run.

The Details of a Gas Engine.

The elements and principal events of gas engine operation have already been cited, and it will be unnecessary to repeat them here. A study in detail of the parts and process that make the gas engine a success will be taken up.

Carburetion.

As previously mentioned, whether gas or liquids are burned in the cylinder, it must be properly prepared first. The liquid must be vaporized and the proper amount of air added to support the combustion. In the gas fuel only the mixture of air need be added. There are numerous devices for receiving the gas, or gasoline or other liquid fuel vaporizing and mixing the vapor with air. A carburetor is a mixer of fuel with air. On many engines the carburetor is called a mixer. See Figure 3.

When using a liquid fuel it must be broken up into fine particles or vaporized. Secondly, it must be thoroughly mixed. Thirdly, the proper amount of air must be added.

The operation. mixing is a simple process. The accompanying illustration will show one way in which it is done. The large pipe is the air inlet, and the small one meeting it at right angles is the fuel supply. When the air is drawn into the cylinder the fuel is drawn into the passageway by the same suction, and as it passes the needle valve it is divided into a thin spray. The air picks it up and carries it into the cylinder. The needle valve not only breaks up the fuel into a spray, but by its adjustment regulates the amount of fuel mixed with the air to make a perfect combustion.

Most of the air passageway is obstructed at the part where the fuel enters, in order to increase the velocity of the air as it passes the fuel inlet, which helps to break it up into a finer spray or vapor.

The liquid fuels which are on the market today do not completely vaporize by the above proces Many methods to accomplish this have been invented and put on the market. Since gasoline has become so high priced, because of its scarcity, the manufacturers, in order to get more of the product have

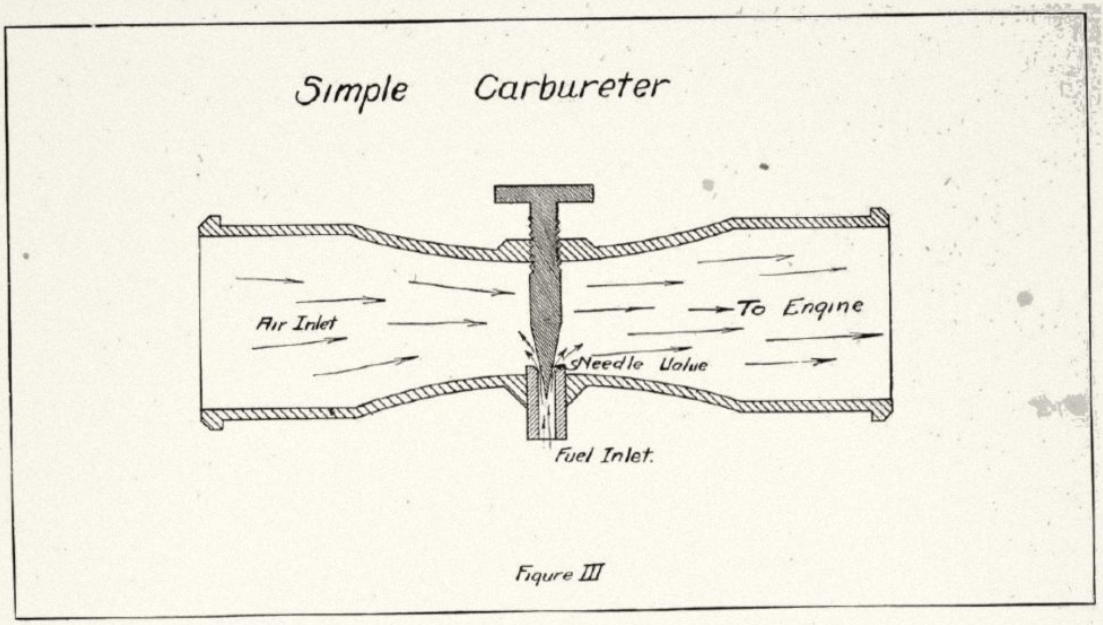

Simple Carbureter
Air Inlet
To Engine
Needle Valve
Fuel Inlet.

Figure III

decreased its quality. This lower grade of gasoline is harder to vaporize than the original gasoline used when automobiles and gas engines first came into use.

Various methods of applying heat to the gas are now on the market and one of the simplest schemes is to take the air which is used in the mixture from off the hot pipes that leave the exhaust chamber. After the engine is running and the exhaust has heated the pipes, this method works all right, but in cold weather it is hard to start such a device until the carburetor and adjacent parts are warm. This can be done with hot cloths or hot water.

Adjusting the Mixture of Fuel and Air.

As the supply of air for the mixture is drawn from the atmosphere, it will be subject to the change of atmospheric conditions.,A change in temperature will change the volume admitted and also the amount of moisture that may be carried by the air. It will also change the vaporizing power as noted above. Therefore carburetors are fitted with certain adjustments to change the proportion of fuel and air to meet atmospheric conditions. A simple carburetor is shown in Figure III.

Also the speed of the engine will vary the suctions so that adjustments must be made automatic to give the proper mixture of fuel during high speed as well as low. The best carburetors on the market will do this.

It is the function of the carburetor or mixer to properly proportion the gases before they are turned into the engine. To regulate the speed of the engine, the governor will open and close the throttle as the speed is being increased or decreased respectively. The throttle is nothing more than a damper valve placed in the passageway between the carburetor and the cylinders. In the hit and miss type of governor, the number of charges taken into the cylinder, is controlled to meet variations in load conditions.

By experiment, it has been found that there is one mixture of air and fuel which is the most economical, i. e., one which will produce perfect combustion and give out the most power. If more fuel is used, the mixture is too rich and will not burn properly, if less fuel is used, the mixture is too lean with practically the same result. Either too much or too little is a waste.

Lean mixtures will cause flames to pass through the air openings, which is known as backfiring. Neither will the engine pick up speed when the mixture is too lean, for it lacks power.

Back-Firing.

Back-firing is caused by a lean mixture as explained above. The fuel charge burns so slowly in the cylinder that when the intake valve opens on the following suction stroke, the gas which is still flaming ignites the new charge, which explodes back through the carburetor; through the air passage.

To remedy this trouble, open the needle valve to give more fuel or cut down the amount of air, which is the same thing. The manufacturer of

every gas engine will give proper instructions for the regulating or adjusting of the carburetor for the conditions which are liable to happen during its life.

Rich Mixture and Its Effect.

When black smoke issues from the exhaust pipe a rich mixture is indicated. An odor of fresh fuel will also be detected in the waste gases. This waste fuel ignites and explodes in the exhaust passages and sometimes blows off the muffler. A rich mixture will also heat up the cylinder quickly, and the ignition points or spark plugs will become covered with carbon deposits in a short time and will make the cylinders miss, a decrease in the amount of power resulting.

To correct this, the process is just opposite to a lean mixture—either close the fuel valve or add more air.

When the mixture is properly proportioned, the exhaust will be clear. Do not mistake the blue smoke coming from the exhaust for a rich mixture, because this is due to over-lubrication. It is a black smoke which indicates too rich a mixture. In kerosene engines a blue smoke may be seen coming from the exhaust at times. This will indicate incomplete combustion or excess lubrication.

Conclusion—The method of the modern manufacturer is to test out the mixing values and carburetors of his special engine and if the fuels which he recommends are used, there should be no trouble in operating the engine. Study carefully the discussion given here, and most cases of trouble will be taken care of with the least expense.

Gas Engine Combustion.

"The ordinary definition of "combustion" means simply burning. The burning of fuel in the gas engine cylinder is the same simple chemical process. It takes place under peculiar conditions. Ordinarily one sees these same fuels burning in the open with excess of air. But actually the fuel is in excess and the burning smokes considerably. In a gas engine cylinder, the mixture of fuel and air is so perfectly made that the result of the combustion is a clear gas and the time of burning is so sudden that it is called an explosion.

Fuels.

All products of petroleum, such as gasoline, kerosene, etc., are composed of the compounds of hydrocarbons which are combinations of carbon and hydrogen., These two elements occur in various proportions depending upon. the specific gravity and the addition of other elements. Atmospheric air is a mixture of, approximately, 23 parts by weight of oxygen and 77 parts by weight of nitrogen. This is a physical mixture.

Carbon and hydrogen as a fuel are in chemical combination and in order for combustion to take place the combination must be broken up. It is partly accomplished by the spray, and partly by intense heat.

THE COMBUSTION.

The air is our chief supply of oxygen. Oxygen is necessary to support combustion. This is why air and the fuel are drawn into the cylinder together. As soon as they are brought into the cylinder, they are compressed, in order to bring the molecules of the oxygen and of the hydrocarbons close together. Then by electric spark or some flame the mixture is ignited and an intense heat occurs as a result of the combustion of all these elements. The carbon and oxygen unite to make carbon dioxide, and the hydrogen unites with the oxygen to make water.

It would be ideal if only oxygen could be added to the fuel mixture, but it is cheaper to get the oxygen from the air, and since the air contains only 23 percent oxygen, a large quantity of it must be used. It has been theoretically determined that 203 cu. ft. of air are required to burn one pound of gasoline. In practice, from 240 to 260 cu. ft. are drawn into the cylinder for the proper combustion of the pound of fuel. It has been said before that each engine has its own peculiarities. The amount of air and gasoline should be mixed according to its demand.

Combustion and Explosion.

The process of combustion in the engine is oftentimes called an explosion. This needs an explanation. It is very near an explosion, and in one sense it can be said that itis. An explosive compound is a mixture of carbon, hydrogen and oxygen chemically, so that there is no necessity for an outside supply of oxygen, that is, all of the combustible parts are contained within the chemical mixture or compound. When set on fire the chemical change takes place without the assistance of any outside influence excepting heat. Now the difference is only this: hydrocarbons of the fuel in the gas engine are mixed physically with the oxygen of the air, while the hydrocarbons in an explosive compound are combined chemically with the oxygen. The explosion is quicker, more certain and reliable. The combustion in the gas engine cylinder is slower!

Timing the Ignition.

The compression does not take place instantly. The ignition of a charge is usually a little bit previous to the time when it is actually needed in order to give the flame a chance to spread throughout the whole mixture. Therefore, the spark is applied to the mixture a little bit before the piston arrives at the compression end of the cylinder. As soon as it has completed its motion, combustion will have been completed, the temperature of the gases raised to the highest point, and the pressure developed so that the piston will be pushed back with maximum speed.

To the human senses it appears that combustion takes place instantaneously. However, with accurate instruments it will be found that this is not true. Some fuels will burn faster than others, and the speed is also affected by the volume of the charge, the temperature, and the perfection of the vaporization and proportioning of the air. Atmospheric changes during the year

will affect these conditions, and make the starting and working of the engine erratic at times.

Cooling the Engine.

Almost constant burning of fuel inside the engine, at a high temperature, heats the cylinder up to two thousand to twenty-five hundred degrees F. Therefore it is necessary to provide some means of carrying off this heat. Water and air are the two common mediums for doing this. If these fail to do their duty the cylinder will heat up, and before the electric spark will have a chance to ignite the fuel charge the heat of compression and the heat of the cylinder is likely to discharge it. This occurrence is called pre-ignition or premature ignition and is recognized by a knocking sound. It may be caused by four other reasons: 1st, compression too high; 2nd, glowing carbon or glowing sharp points on the cylinder wall; 3rd, uncertain ignition, and 4th, pockets of burning gas in the combustion chamber. These defects will be discussed later.

The Source of Power in Gas Engines.

The ideal fuel for gas engines is a liquid or solid fuel, which can be easily vaporized and mixed perfectly with air to form a combustible vapor. In burning it should leave no carbon deposit, nor should it give off any smoke or foul gas. This kind of fuel has not been found in practice, although high grades of gasoline approach it very nearly.

Kinds of Fuel.

The total number of fuels that are available and profitable to use is so large that only the most important and common ones will be mentioned. Standing first in America are the petroleum products, such as benzine, naphtha, gasoline, kerosene, distillate, and crude oil. In the next important group are natural gas, illuminating gas, power or producer gas, alcohol, and wood oil.

The petroleum products are largely obtained from North America. There are large deposits in Russia, and many other countries, but the American fields easily supply half the world. There are two kinds of petroleum by-products in the United States, and in general the Mississippi divides the fields containing the two products. The Eastern field contains the products having a paraffin base, and the Western field contains a product having an asphalt base. Paraffin base means the character of the product left after the gasoline, naphtha, kerosene, and distillate have been driven off by heating. Along the Mississippi Valley the two kinds of oil are mixed.

Composition of Crude Petroleum.

To the user of a gas engine, the exact composition of crude oil is not of much interest except in its broad economic sense. It will be noticed from what follows that certain parts of the crude oil are more valuable than others. The simple conclusion would be for an engine which could use the most available product to gain by getting the cheap fuel. In Figure 4 is graphically shown the proportion of gasoline, distillate, lubricating oils, asphalt residue and waste from the Western petroleum. Gasoline is seen to be a very small part, and it has therefore become high priced as the demand for it increased. Kerosene, which is about one-

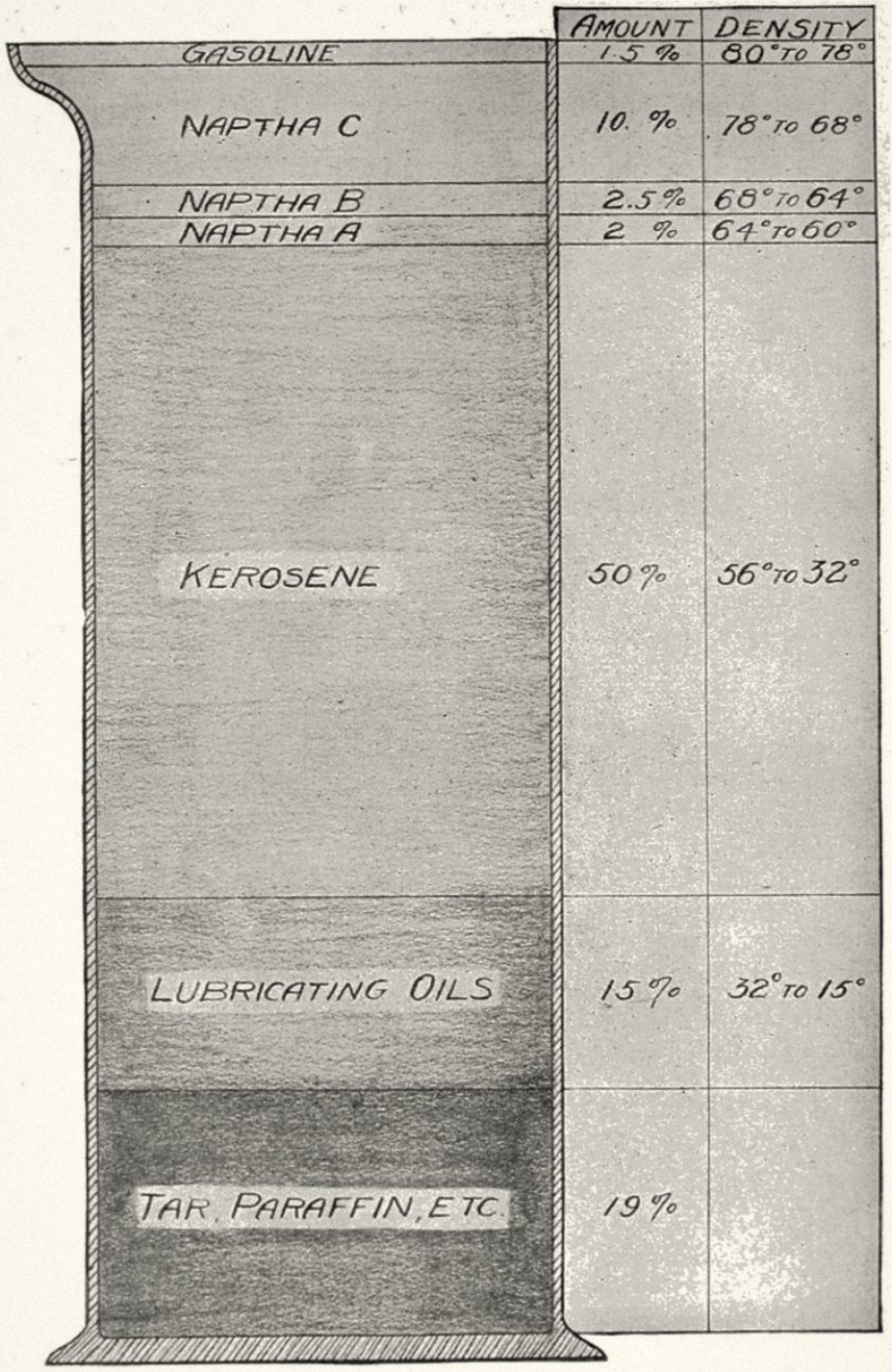

AMOUNT
DENSITY
GASOLINE
1.5 %
80° TO 78°
NAPTHA C
10. %
78° TO 68°
NAPTHA B
2.5 %
68° TO 64°
NAPTHA A
2 %
64° TO 60°
KEROSENE
50 %
56° TO 32°
LUBRICATING OILS
15 %
32° TO 15°
TAR, PARAFFIN, ETC.
19 %

Figure 4

half of the total product, in view of its availability, is cheaper than gasoline. This will help to explain some of the differences between gasoline and kerosene on the market today.

Separation of Petroleum Products.

It will be interesting to know how these products are separated. The process is known as "fractional distillation." Crude oil as it comes from the earth is a thick, heavy black liquid. This liquid is put into large vessels and heat is applied to the exterior, the temperature being maintained constant at various points along the line. That is, the whole mass will be heated up to the temperature at which the gasoline vaporizes, and will be held at that temperature until all of the gasoline has passed off as a gas. It is led through a series of tubes and cooled and condensed until it will flow into another vessel. Then the mass is heated again to a higher temperature, at which kerosene will vaporize, and kept at the temperature for some time, and the kerosene is taken off in the same way as the gasoline, and so on for the lubricating oils. The paraffin base of the residue is used directly in many commercial fields, or it may be further distilled, and produce vaseline and a number of other well known petroleum products.

The original gasoline of a few years ago was 3 percent. The demand has become so great for fuel that some of the higher grades of kerosene have been added to the gasoline, thus decreasing the quality of gasoline, but increasing the amount available. Recent inventions have brought in new methods of "cracking" some of the lower distillates and producing gasoline. This will help to save the gasoline as a fuel of the future. Perhaps further progress will soon be made along that line and this most wonder-fuel will be saved for us through many years to come.

Petroleum Products Classified.

Petroleum products can not be classified by the temperatures at which they are distilled. On the market they are known by their weight or specific gravity. The following diagram will show the correct approximate weight of petroleum products and the various distillates, and will explain what is meant on the market by tests of 70 or 80, etc.:

The Baume Scale.

The Baume Scale is of French origin and was invented to make a more convenient form. As an example, suppose that a gasoline had a specific gravity of.9 and also has a Baume reading of 75 degrees or 75 degrees Baume. The terms Baume gravity and degrees Baume are the same. To convert Baume to specific gravity or specific gravity to Baume the two following formulas may be used:

$$\text{Baume gravity} = \frac{140}{\text{Specific gravity}} - 130$$

$$\text{Specific gravity} = \frac{140}{\text{Baume gravity} + 130}$$

CORRECT APPROXIMATE WEIGHT OF PETROLEUM PRODUCTS.

Product	Degrees Baume	Specific Gravity	Pounds in One Gallon
Petroleum ether	95	.6222	5.18
	90	.6363	5.30
	85	.6511	5.42
Gasoline	85	.6511	5.42
	80	.6666	5.55
	75	.6829	5.69
	70	.7000	5.83
	69	.7035	5.86
	68	.7070	5.89
	67	.7106	5.92
	66	.7142	5.95
	65	.7179	5.98
	64	.7216	6.01
	63	.7253	6.04
	62	.7290	6.07
Naphtha	62	.7290	6.07
	61	.7329	6.11
	60	.7368	6.14
	59	.7407	6.17
	58	.7446	6.20
	57	.7486	6.24
	56	.7526	6.27
	55	.7567	6.30
	54	.7608	6.34
	53	.7650	6.37
	52	.7692	6.41
	51	.7734	6.44
	50	.7777	6.48
	49	.7821	6.52
	48	.7865	6.55
	47	.7909	6.59
	46	.7954	6.63
Kerosene	46	.7594	6.63
	45	.8000	6.66
	44	.8040	6.70
	43	.8092	6.74
	42	.8139	6.78
Distillate	42	.8139	6.78
	41	.8187	6.82
	40	.8235	6.86
	39	.8284	6.90
	38	.8333	6.94
	37	.8383	6.98
	36	.8433	7.03
	35	.8484	7.07
	34	.8536	7.11
	33	.8588	7.15
	32	.8641	7.20
	31	.8695	7.24
	30	.8750	7.29
	29	.8805	7.34
	28	.8860	7.38

Petroleum Fuels Now on the Market.

As mentioned previously, the great demand for gasoline in the past decade has forced the manufacturers to decrease in quality in order to get more quantity. It is quite common now to get gasoline as low as 60 Baume and in some extreme cases they are listing it as low as 56 degrees, with kerosene as high as 48 degrees Baume. This fuel can be used in the small farm engines just as well as the high grade fuels with the exception that the starting is a little more difficult. If instructions to warm the fuel and engine are followed strictly, there will be no trouble in using them.

Heating Value of Petroleum Products.

Although the commercial value of petroleum products is easily determined by the degrees Baume, it is another one of the properties which really decides the matter. Fortunately the higher degrees Baume correspond exactly with the higher heat values of the fuels.. Heat is measured in “British thermal units” (B. t. u.). A British thermal unit is understood to be the quantity of heat required to raise the temperature of one pound of pure water, one degree Fahrenheit Below will be given a table showing the B. t. u. value of the petroleum products before listed.

Although the number of heat units vary in the different products, the available power from each engine using that particular product seems to be about the same. The efficiency of the machine depends more upon the design of engine than upon the kind of fuel.

Power Rating of the Gas Engine.

Manufacturers of tools for any industry have certain terms which have special meanings, to describe the properties of their products. It is difficult for them to use any other terms, and even in talking to a layman, these terms become so familiar to the manufacturer that he feels because they are so familiar to him they must be to every one else. When he comes in contact with a layman, who has not encountered these terms, there is necessarily some confusion. The layman can see by his manner that he is expected to know what the terms mean, and rather than embarass his friend he will give his attention and assent to many things which he does not understand. Such terms as “power rating,” “piston displacement,” “Horse Power,” “Brake Horse Power,” “Draw Bar Horse Power,” “Foot-pound,” “work,” “power,” etc., will be defined here in a simple way and thus bring about an understanding between two friends.

Power rating is a term used to express the strength of a gas engine, or the amount of work it will do in a given time.

Displacement in cubic inches, is that volume of the cylinder through which the piston passes. It is the same as the volume of the cylinder whose height is the length of the stroke, whose diameter is the diameter of the piston. Automobile engines which are qualifying for races must not allow piston displacement to be over a certain number of cubic inches, so that the best car will be the one which will make the best use of the fuel in a given space. Competition, therefore, is between quality of design and not in size of car or engine.

Power.

Power is defined as the amount of work that can be done in a unit of time.

Work is a unit produced by a force acting through a distance and is measured in foot-pounds. That is, a foot-pound is the amount of work required to raise one pound a foot high,

Remember that work and power are not the same. Work does not take into consideration the time element, but power does.

The unit of measurement of power is defined by James Watt as the rate of doing 33,000 foot-pounds per minute.

Indicator or Indicated, and Drawbar Horse Power.

Mechanical engineers have a method of measuring the amount of power expended by the burning gases in the cylinder. The instrument they used is called the “Indicator” and the horse power determined is the indicated horse power.

Horse power referred to above is not that which is available for use, so that power is measured again at the fly-wheel. The power developed by the engine and transmitted to the belt is measured by means of a Prony Brake. The Prony Brake is clamped on the fly-wheel much the same as shown in the figure and the bar weighted so as to give the engine a uniform motion consuming the maximum amount of fuel. The calculations are then made giving the amount of horse power delivered.

The difference between the last two horse powers will be the mechanical friction of the engine parts.

In motor vehicles, such as automobiles, tractors, etc., another kind of horse power is used. This is called the “draw bar horse power,” and it measures the pulling power of a tractor or motor vehicle at the draw bar. Usually the draw bar horse power is about half of the brake horse power, that is, half the power is lost in the transmission and friction of gears and bearings.

Calculating the Engine Horse Power.

The horse power of engines may be determined by formula, but this method is inexact unless the exact quality of the fuel is known. However, for ordinary use it will give an approximate value. The formula for gasoline burning engines is as follows:

$$\text{Horsepower} = \frac{DLN}{16{,}400} - 0.5$$

and for kerosene burning engines is

$$\text{Horsepower} = \frac{D^2LN}{21{,}875} - 0.75$$

D is the diameter of the cylinder; L is the length of the stroke; N is the number of revolutions per minute. Diameter and length are measured in inches. This will give the power for a single cylinder engine. When more than one cylinder is used, multiply the results by the number of cylinders.

The Best Way to Get Horse Power.

The only sure way to know the horse power of the engine is to test it. The Prony Brake or drawbar spring either can be used to determine the respective kind of horse power.

Formulas will be found unreliable, since all they can do is to give the dimensions of the cylinder, and some constant representing the quality of the fuel. One automobile concern has been able to almost double the power of its engines without increasing the size. This is accomplished by greater refinement in design.

Brake Test for Gasoline Engines.

The farmer often would like to know whether the manufacturer has given him all that he has paid for. This he can easily determine himself by the following method. Construct a Prony Brake as shown in Figure 5. This can be done by anyone with four pieces of 2-inch lumber, a piece of strap iron, a platform scale, nails, screws, blocks, etc., as shown. The friction blocks that are set against the wheel should be of soft wood and a little wider than the wheel. It is convenient to notch them so they will not slip off sideways.

Operation.

Place the brake on the wheel as shown, resting the shoe on the platform scale and with the wing nut on the strap loose, to balance the platform scale. Start the engine while the brake is loose. Slowly tighten up the wing nut, at the same time putting on more power. Keep the platform scale balanced. When the full amount of fuel has been admitted, adjust the wing nut so that uniform speed will be maintained. With a speedometer or counter determine the number of revolutions per minute. By using the following formula the brake horse power will be determined. Compare this with the manufacturer's statement of the brake horse power.

$$\text{H. P.} = \frac{(W)\ (A)\ (N)}{5252}$$

Where W = Weight registered on scales. A Lever arm (see figure). N Number of revolutions per minute.

Follow the directions carefully. If one has never made the test before he should proceed with caution. Be sure that the brake is put on the fly-wheel so that it turns in the direction indicated by the arrow. To make an accurate reading follow the formula.

If the engine should test higher than it is rated of course this will please. Tt should not be run at its maximum rating, however. Usually the manufacturers have tested out the engine and considered the rating at which it will be the most economical. Engines should be run at nearly their full horse power but in no case should they be overloaded.

The Need for Testing the Engine.

It is not so much to compare the rated horse power with the actual as it is a matter of maintenance that the brake test should be used. If the owner has

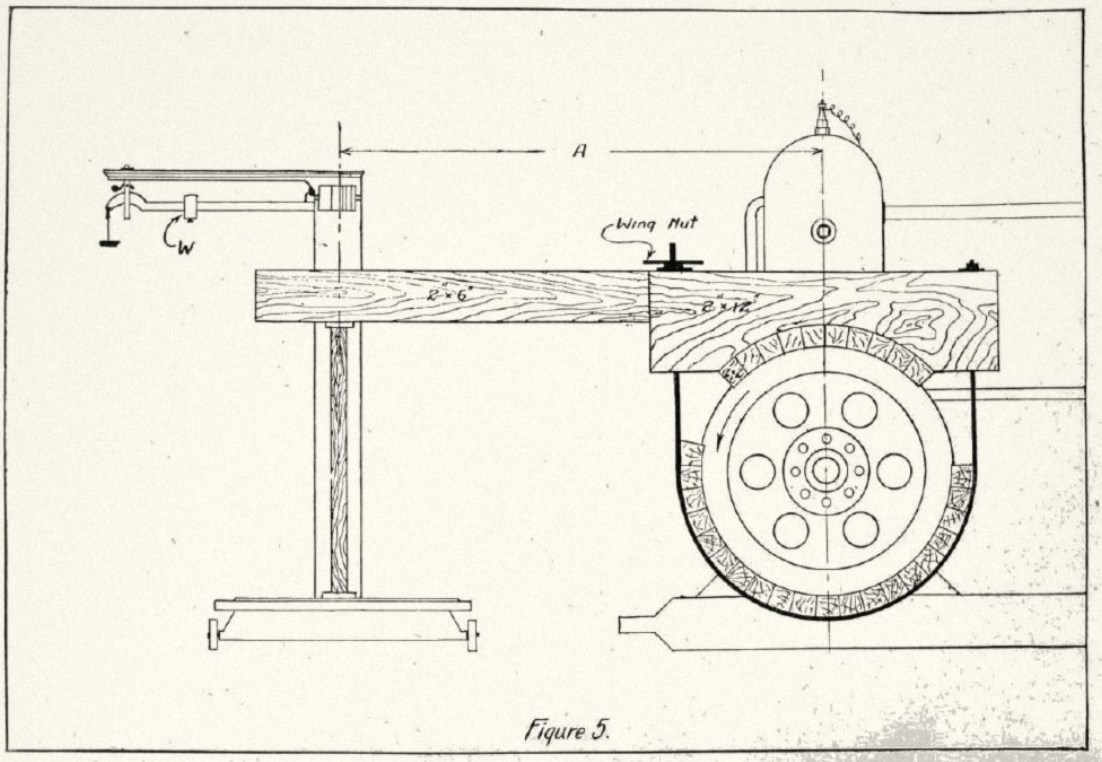


Figure 5.

tested the engine when it is new, then once or twice a year repeating the process he will always have an idea as to the condition of the engine. He will know whether it is losing strength or gaining. In case he finds the engine is losing its power he should investigate immediately. (See chapter on Troubles.)

Economy Test.

Manufacturers will not only rate the horse power of the engines, but also say on what fuel the horse power can be maintained for a certain number of hours.

Proceed as follows: After determining the brake horse power test, set the scale beam at the rated horse power loaded, and by adjusting the wing nut keep the scale beam balanced and run the engine for half an hour or an hour. At the end of this period of preliminary running, if everything is found to be uniform, measure the amount of fuel by weighing, and proceed with the test, noting the time it takes to uSe up a weighed amount of fuel. The amount of fuel used may be that which the manufacturer claims will do the work in the given time. This will be found the simplest way to run the economy test.

The Power House or Shop on the Farm.

Economical use of power on the farm is obtained mainly by concentrating it in one plant as near as possible. Of course there are some activities that must be handled by individual plants, such as a tractor for plowing, hauling, etc. This chapter will be given over to things which can be done mainly by a stationary plant. The investment in the engine is oftentimes a burden, and unless it is kept at work it will not pay. An idle farm hand is certainly a losing proposition. An idle engine will of course give similar results.

Mistakes are often made in applying farm engines to work which is only done once or twice a year and over a short period of time, and leaving such work as is done every day to man power. This is a very serious mistake. Such work as pumping water, feed grinding, washing, weighing, churning, cream separating, dish-washing, milking, etc., should be the first ones to receive consideration. After the daily routine has been taken care of the special uses may be taken up in order of their importance.

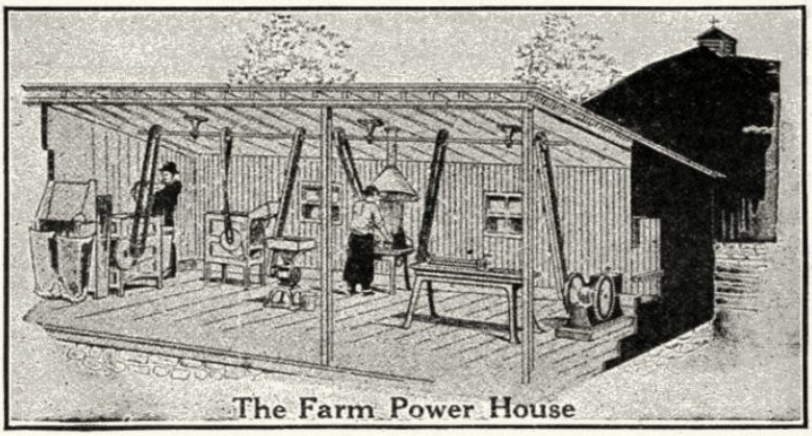
One of the first requirements for the proper use of an engine on the farm is a power house. In one corner of this power house should be a small room closed off with a dust-proof partition, in which the engine shall be placed. It should be placed on a solid foundation, with the fuel tank, water tank, etc., placed in the most advantageous position. The power may be transmitted by means of a line shaft or by an electric motor generator set much cheaper than the engine could be moved around from place to place as needed. The building should be made long, and narrow if necessary, with a line shaft running the full length as shown in cut. Partitions can be placed between different types of machines. Some are necessarily dusty, and should be separated from the rest.

Care.

Too much cannot be said in regard to the care of the machinery, especially the gas engine. There is nothing that will depreciate when exposed to weather more than a gas engine. It would be a better investment in every way to leave the gas engine in a well protected building and transmit the power by line shaft and belting-or by electrical arrangement, than to continually move the engine and necessarily expose it to the elements.

Location of the Power House.

If a great amount of the power is used in connection with the barn, it should be located nearest the barn. If a connection with the house is impossible with



The Farm Power House

a line shaft, and the expense too great to do it electrically, another small engine is usually the most ‘economical way for handling that trouble. The power to operate washers, sewing machines, etc., in the house are very small compared with that necessary for feed grinders, milking machines, buzz saws, etc.

Design of the Power House.

It is often necessary to build an entire new house, and this is the most advisable if funds will permit. In the beginning a section of the barn or wood shed might be used. After a small concrete mixer is installed the blocks can be made and as the farmer is not very busy at certain times of the year he should build the shop and move in at a convenient time. Considerable care should be given to the design of such a house, because changes are expensive. As has been said before, the building should be so made and arranged as to separate the various kinds of uses of the power and to fully protect the engine in a separate room. Diagrams are shown giving the ideal conditions. Of course these cannot be maintained in every place, but it will give some idea as to the best, and towards what one should work.

Equipment of the Power House.

After a satisfactory house has been built it should not be expensive to add the equipment and connect up the power with the machines to be used. A good buyer will look around the junk man's iron and steel pile and oftentimes find pieces of shafting and pulleys which he can get for a very small price and which will answer the purpose satisfactorily.

Shafting.

Cold rolled steel shafting is the best for this purpose. In the table below is given the size of shaft, revolutions per minute, and horse power generated or transmitted. Following this table will be found another table showing the amount of horse power the different widths and thicknesses of belts will transmit with speed of a hundred feet of belt per minute.

HORSE POWER OF SHAFTS FOR GIVEN DIAMETER AND SPEED.

Diameter of Shaft Inches	Revolutions per Minute									
	100	125	150	175	200	225	250	300	350	400
$1\frac{3}{16}$	2.4	3.0	3.6	4.2	4.4	5.4	6.0	7.2	8.4	9.6
$1\frac{7}{16}$	4.3	5.4	6.5	7.6	8.6	9.8	10.8	13.0	15.2	17.2
$1\frac{11}{16}$	6.5	8.0	9.7	11.2	13.0	14.6	16.0	19.4	22.4	26.0
$1\frac{15}{16}$	10.0	12.5	15.0	17.5	20.0	22.5	25.0	30.0	35.0	40.0
$2\frac{3}{16}$	14.0	17.8	21.0	24.5	28.0	31.5	35.6	42.0	49.0	56.0

HORSE POWER BELTING WILL TRANSMIT.

Width of Belt Inches	H. P. per 100 Feet Belt-Speed		Width of Belt Inches	H. P. per 100 Feet Belt-Speed		Width of Belt Inches	H. P. per 100 Feet Belt-Speed	
	Single Belt	Double Belt		Single Belt	Double Belt		Single Belt	Double Belt
1	.09	.18	9	.82	1.64	20	1.82	3.64
2	.18	.36	10	.91	1.82	22	2.00	4.00
3	.27	.55	11	1.00	2.00	24	2.18	4.36
4	.36	.73	12	1.09	2.18	28	2.55	5.09
5	.45	.91	14	1.27	2.55	32	2.91	5.82
6	.55	1.09	16	1.45	2.91	36	3.27	6.55
7	.64	1.27	18	1.64	3.27	40	3.64	7.27
8	.73	1.46						

Pulleys.

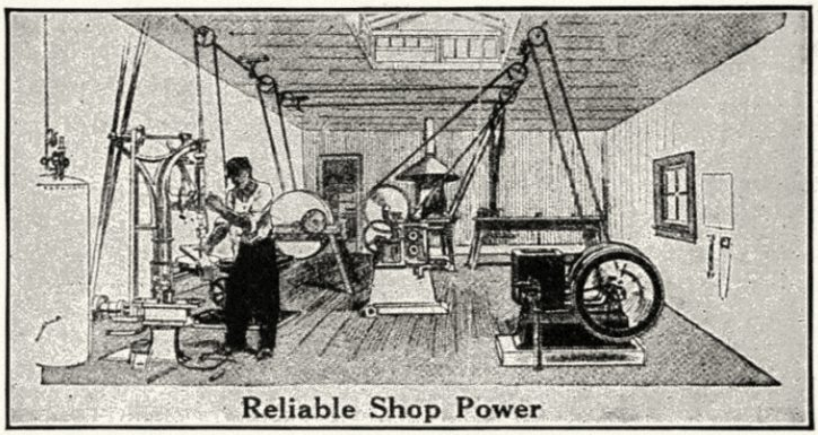
The local hardware man or implement dealer will be able to supply all the materials necessary from stock. If not you can easily order them. Use split wood pulleys, as they are most convenient to take off and put on when changes are necessary. However, the driving pulleys and any such pulleys as may be needed near the end of the shaft could be made of cast iron or steel, as these will be easily removable from the end of the shaft, and be more permanent than the wood.

Spacing the Hangers.

The hangers of the shaft should be spaced close enough to prevent any sagging, and especially should hangers be near heavy pulleys or pulleys with heavy belts carrying big loads. An extra hanger is of small importance in cost and will be very valuable in maintaining the shaft and bearing because if a shaft is allowed to bend too much it burns away the bearing and will ruin the shaft. An extra hanger or two will always prevent this.

Machines and Line Shaft.

The machines should always be placed with respect to the line shaft so that the belting will make an angle of about forty-five degrees with the horizontal. The distance from the pulley to the machine should be from eight to ten times the diameter of the largest pulley.



Set machine so that the center lines of the shaft of the machine and the line shaft are parallel so that the center lines of the pulleys will coincide.

The Engine and the Machines.

If possible place the largest machines or those requiring the most power near the engine. In transmitting the power the line shaft is twisted. The work of twisting the shaft takes up some of the power of the engine, and consequently, the shorter the shaft the less power will be lost this way.

Managing the Power Plant.

After the power house, machines, and engine have been placed, another very vital problem arises. The engine of course is large enough to drive the largest machines and may be made large enough to drive more than one of the large machines, if it is necessary during a day’s use to put two machines in operation at the same time. Bear this in mind, that the engine should be run at its rated

capacity if possible. Then it will consume the least amount of fuel per horse power hour. Therefore arrange to have one or two of the larger machines and several of the smaller machines running at the same time. Farm work may be managed to bring about this result. Also the caretaker of the machines and engine will find it possible to take care of several machines as well as one. Then the engine and one machine would not require all of his time and what was not necessary would not be lost.

A speedometer or counter should be one of the tools about the power house. This is a convenient piece of apparatus shown in Figure 6.

Pulleys, Speeds and Sizes.

The calculation of pulleys, speeds and sizes is a very simple matter after the speed of the engine and the speed of the machines are shown. There are two kinds of pulleys, the driving pulley and the driven pulley. The driving pulley is the one at the engine or source of power and the driven pulley is at the machine or where the power is being used. The speed of the pulley will be represented in revolutions per minute (RPM). The diameter of the pulley will be called "D." When it is the driven pulley to have a subscript (m) and when it is the driving pulley it will have the subscript (p); (m) standing for machine and (p) for power. The speed will be indicated by (RPM), subscript used Fig. 6 the same as for diameter of pulleys. Four formule will be given as follows:



1. $D_p = D_m \frac{(RPM)_m}{(RPM)_p}$.

2. $(RPM)_p = D_m \frac{(RPM)_m}{D_p}$.

3. $D_m = D_p \frac{(RPM)_p}{(RPM)_m}$.

4. $(RPM)_m = \frac{(RPM)_p}{D_m}$.

TROUBLES.

Gas engines have a reputation for causing trouble, which they are fast overcoming. Many users have little or no complaint to make. Many instruction books list the parts causing trouble instead of the trouble. In this discussion the operator will observe the symptoms of the trouble and find the remedy listed under that heading.

Gas Engine Troubles.

1. Engine will not crank—engine stuck. 2. Engine will not start. 3. Low power—loss of power. 4. Misfiring. 5. Misfiring in one cylinder. 6. Sudden stop. %. Back-firing. 8. Irregular running. 9. Overheating in cylinder. 10.

Pre-ignition or pounding in cylinder. 11. Crank case explosions—two-cycle. 12. Smoke. 13. Engine gradually slows down and stops. 14. Excessive vibration. 15. Cam shaft rattles. 16. Wheezing scraping sounds. 17. Knocking and pounding—regular. 18. Irregular knocks. 19. Speed variation.

The above troubles will be taken up in detail and the cause or causes will be given and remedies for each.

It is suggested that the operator of a motor first look over the list above and find which one of the main troubles he is having, then turn the pages ahead which take up such trouble in detail. Under each of the main headings will be found the causes with their remedies.

Troubles Analyzed.

Hard to Start.—The engine may refuse to start for the following reasons: hot bearings, overheated piston, water frozen piston cylinder, bolts rubbing on ase or oil shields, friction clutch holding load on engine, broken gears and wedges, water in cylinder, obstacles blocking wheel or gears, broken crank shaft, and dry bearing or rusted piston.

Hot bearings may be caused by tightness, poor lubrication, scored bearing surfaces, bent or sprung shaft, poor alignment, or shaft out of round. If bearings are tight, loosen them. If they are poorly lubricated, put on some more oil. If the

earing surfaces are scored, they will need to be put in a machine and machined until the scores are removed. It might be necessary to put in new bushings. If the shaft is bent or sprung, it must be taken to a machinist and straightened. The poor alignment may be corrected by adjusting the shaft and bearings. If these are not adjustable, take them to a machinist for correction. If the main shaft is out of round, it should be given to a machinist and turned to a true surface and new

earing fitted.

f the piston is stuck due to overheating, first cool the cylinder, then add more oil and decrease the load. Arrange to have the cooling system supply a larger quantity of cold water to prevent happening again.

f the cylinder leaks and the water is frozen the best thing to do is to pour hot water in the radiator or the water cooling tank to warm up the whole machine.

Bolts rubbing on the base or oil shields. The bolts will have to be filed down so as to miss the base, or the oil shield may be bent out of place.

If the friction clutch is holding the engine, it should be uncovered and loosened.

f broken gears give trouble, they will have to be taken off and replaced.

f there is any obstacle or locking of wheels or gears this can be sometimes determined without taking off the gear casing from the fact that the gears will turn one way and will not turn the other. The gear casing may be taken off and the gears inspected. Look for other troubles first.

Open the crank case and see if the crank shaft is all right, it may be broken. Tn that case a new crank shaft will have to be secured.

f the engine has been standing a long time, the bearings may be dry or rusted and a thorough dismantling and cleaning up will be necessary and plenty of oil

Starting Troubles—The engine may refuse to start for the following reasons: the fuel valve closed in tank, no gasoline in the tank, gas shut off at

meter, battery or magneto switch open, broken or disconnected magneto wire, dirty electrodes, broken ignitor springs, weak batteries, magneto not generating, foul spark plugs, short circuit, defective spark coil, defective timer, storage batteries, no compression indicator, carburetor trouble due to poor mixture, cold weather carburetor trouble, high altitude, air leaks in cylinder, vibrator on high tension spark coil may be out of action.

If the engine refuses to start, investigate the fuel supply first, then the magneto and battery switch. If these are found all right, see if there is a broken connection, or dirty electrode, or a broken ignitor spring. Test the batteries and magneto for weakness. Take out the spark plug and see if it is clean; there may be a speck of carbon between the two points. Be sure that the electric wires are not short circuited or crossed in some way. The spark coil may be defective, and if it is, it should be taken to an electrician for repairs. Open the timer and see if the contacts are all right. The piston may leak and give low compression. Due to a sudden change in weather or to tinkering, the carburetor may be improperly adjusted and the mixture not properly made. If there are any leaks in the cylinder between the carburetor and the cylinder, these should be closed up, as a certain amount of air will get in. Watch the vibrator. It may be out of order. When the engine is turned over see that the vibrator buzzes.

No Power or Loss of Power.—Power loss trouble may be brought about by the fuel valve being partly closed, the air damper closed, compression release cam being left in “starting” position, throttle left in “starting” position, retarded sparks reducing power, advanced spark causing excessive loss of power and pounding, weak batteries, vibrator adjustment, spark plugs foul, foul ignitor on make and break system, defective timer, clogged muffler, long exhaust pipe or exhaust has short bend, magneto troubles, lack of oil, hot bearings, crank case leakage on two-cycle engines, and carburetor troubles.

Fuel valves, dampers, etc., may jar shut or out of position. These should be inspected when the power seems to be varying. Note and see if all the adjustments have been changed from starting position to running position. A retarded spark will reduce the power and an advanced spark will cause pounding and also reduction of power. Weak batteries will give a poor spark and reduce the power. Test the vibrator adjustment and see that it is working. Investigate the spark plugs and ignitors and see if they are fouled. The timer may be out of adjustment or con+ tacts not clean.

If the muffler is clogged up with soot or if the exhaust pipe has too many bends, it will reduce power. Test the magneto. Look at the supply of oil. If the bearings are hot, cool them with cold water and add oil. If the crank case leaks in a two-cycle engine, it must be closed. Inspect the carburetor and see if its adjustments are O. K.

Tf there is a leakage in the compression, it may be found by turning the engine over dead center on the compression stroke. If it passes easy, there is a leak. This should be investigated and stopped up by new rings or machining.

Too cold water on the water jacket will reduce the power. For gasoline engines the water should be about 160 degrees, and for kerosene 200 degrees F.

The valves should be inspected for timing.

Warm air intake will reduce the mixture in the cylinder, and thus reduce the power.

High altitudes will reduce the output of engines.

Worn cams, rollers, and timing gears will change the timing and reduce the power. When the valve opening is too small, it will cause a back pressure and loss of power.

If the spring is too stiff on the intake valve, not enough fuel will get in.

Valves may be stuck in the guides, especially if the oiling is not well done. The magneto may be out of time and the valve gear worn.

Misfiring—Misfiring may be caused by loose wires or dirty connections, swinging ground wire caused by poor insulation, a broken wire, weak or exhausted batteries, poorly adjusted vibrator on high tension system, foul spark plugs, dirty electrodes, moisture causing short circuit, magneto out of order, powerless sometimes caused from misfiring, defective or short circuited spark coils, defective timer, run down storage batteries, water in gasoline, worn valve gear, leaky exhause valve, poor mixture of fuel and air, air leaks between carburetor and cylinder, valves out of time, leaky automatic intake valve, compression leaks, empty fuel tanks, spark gap too large in spark plugs.

The above causes for misfiring suggest their own remedy except in a few cases. The adjustment of the vibrator may be done while the engine is running. The defects may be determined by listening to the buzz when the engine is stopped on the firing position. If the buzz is uniform and continuous the vibrator is probably all right. If not it must be adjusted. Foul or dirty parts must be cleaned. By inspecting the timer, note particularly that the moving parts are well cleaned and that they have a good contact at every turn. In case of water in the gasoline, the tank should be emptied, the carburetor emptied and a complete new supply instilled. Leaky exhaust valves can be determined by turning' the engine over on compression. If the compression is weak and there is a hissing sound in the exhaust valve, it indicates a leak. This may be corrected by cleaning but as a last resort they can be reground with emery. Air leaks between the carburetor and cylinder may be determined by listening closely to the intake valve and intake manifold. To correct, the manifold should be taken off and repacked. If the valves are out of time, the gears operating the cams will have to be taken off and readjusted. Look at the spark plug and see if the spark gap is too large. This can be determined by a test.

Misfiring in Cylinders.—Misfiring in one cylinder may be caused by this cylinder having a heavier carbon dcposit than any other, or one cylinder may have a leak, or it may be out of time, or a poor contact in the timer, or it may have a loose wire lead to the misfiring cylinder, or a sooted plug, or the magneto distributor may be fouled with dust, or one vibrator may be stuck. All of the above causes are practically self-explanatory. Where there are poor contacts these should be corrected by piping or cleaning. Where the wires are loose, they should be tightened.

Sudden Stopping—Sudden stopping is caused when the ignition switch is jarred open, the fuel exhausted in the tank, wires broken, loose connections of the wires, carburetor nozzle clogged with dirt, fuel pipe clogged leading to the carburetor, timer broken, or defective magneto, hot bearings bind the shaft, or defective ignitor, or water in the gasoline, or hot cylinder when the piston is seized, no oil, or a poor mixture of oil, it is hard to find such troubles as open switches,

broken wires or connections, etc., except by opening the casings or protecting coverings and inspecting. The first thing of course is to investigate the control switches and fuel connections. Try to turn the engine over and if it will not turn, the trouble is probably a hot bearing or a hot cylinder. These should be cooled with cold water and properly oiled and if any damage has been done to them it should be corrected before using again.

Back-firing.—Back-firing is caused by a poor mixture due to carburetor adjustment, retarded spark, clogged carburetor nozzle or fuel pipe, leaky inlet valve on engine, air leaks in cylinder, or intake valve, wide open throttle at full load, on low speed may be caused by admitting too much air to the carburetor, valves out of time, or a defective timer on high tension system, and by weak batteries. When troubled with back-firing first try enriching the mixture by cutting off air supply and adding more fuel. If this corrects the trouble, a poor mixture was the cause. Try advancing the spark. If neither of these will cure the trouble, investigate the fuel supply to the carburetor. Test for leaks by putting on compression. If none of these will answer inspect the timing system and see if the timing gear has been placed correctly and last, test the batteries.

Irregular Running.—Irregular running results from broken wire, dirty timer, sticking coil vibrator high tension system, worn make and break mechanism, loose timer control rods, water in gasoline, clogged carburetor nozzle, weak exhaust valve spring or broken springs, air leaks between carburetor and cylinder, worn cams or cam shaft on multi-cylinder engine as well as a twisted shaft or loose gears. The above causes are mostly due to worn parts or negligence. These can be corrected by thorough inspection and overhauling of the engine.

Overheating in the Cylinder.—Overheating in the cylinder will result from a retarded spark, too rich a mixture, lack of oil in the cylinder, poor water circulation due to the jacket, poor compression, insufficient valve lift, clogged exhaust pipe, clogged muffler, clogged radiator, defective circulating pump, tight piston, lime deposits in cylinder, overload on the engine, and a closed water supply valve.

What to Do With an Overheated Cylinder.—Advance the spark, and if this does not remedy it, try enriching the mixture. Next inspect the oil in the cylinder. Inspect the pump and water circulation around the jacket, and try the compression. Then open the valve edge and see if the lift is sufficient. Next inspect the exhaust pipe, muffler and radiator. If the engine will not turn over, the piston is probably tight. Cool the piston while the engine is still hot and this will loosen it. If none of these will do, probably the engine is overloaded. Inspect the water supply valve.

Pre-ignition or Deep Pounding in the Cylinder.—Pre-ignition or pounding in the cylinder is caused by too much lubricating oil forming a deposit, a rich mixture forming a deposit, overheated cylinders especially in air-cooled engines, sharp edges in combustion chamber, deposit in cylinders, and deposits in kerosene engines are usually formed by running with a cold cylinder, or by having the intake air too cold.

What to Do.—Open the cylinder and see if there are any deposits. If there are, use the methods already described to clean them. If there are any

sharp edges in the combustion chamber, file them off. They may be left to grow and cause pre-ignition.

Crank Case Explosions in Two-Cycle Engines.—This will be caused by retarded spark or a poor mixture due to carburetor adjustment. Test both of these and see if the trouble is not eliminated.

Smoke.—Black smoke is caused by too much gasoline or too rich a mixture, light colored smoke is caused by an excess of oil fed to the cylinder. Either of these can be remedied by taking care of the carburetor or the oil supply.

Engine Gradually Slows Down and Stops.—This may be caused by weak or exhausted batteries, poor mixture due to carburetor adjustment, overload on the engine, magneto slipping or governor out of order, or overheated bearings.

Test your batteries. Enrich the mixture. If this does not help, take off some of the load, then try the magneto and see if it is slipping and put hand on the bearings and see if they are hot.

Excessive Vibration—Excessive vibration may be caused by poorly balanced engine crank shaft, twisted cam shaft may change valve timing and cause an uneven application of power, or an uneven wear on the cams or push rods may give an uneven application of power, or a defective timer may fire the cylinders at unequal periods, or the carburetor not situated at equal distances from the cylinders may be the cause of the vibration. To correct this inspect the crank shaft and cam shaft especially as to wear. A machinist only can remedy the wear by machining and rebushing. If none of these are the trouble, note the position of the carburetor and change if necessary.

Cam Shaft Rattle—The cam shaft rattles because of retarded spark, or loose cam shaft bearings or gears, or loose cam rollers or pins. To remedy this, advance the spark, tighten up cam shaft bearings or cam rollers or pins.

Wheezing and Scraping Sound.—A wheezing and scraping sound will be caused by broken piston rings, dry piston due to lack of oil, vibration of auxiliary air valve on carburetor, tight piston, overheated cylinders, or fly-wheel scraping on metal shields. When a wheezing or scraping sound is heard, the cylinder should be opened and examined well for broken piston rings or lack of oil. Also see discussion for overheated cylinders. See if metal guards and shield are not rubbing on some moving part.

Knocking and Pounding—Regular—Knocking and pounding will be caused by ignition too far advanced, pre-ignition, overheated cylinders, loose bearings, loose piston rod bearing, loose fly-wheel, loose counterweights, end play in crank shaft, broken valve stem, broken circulating pump, and wear on cam" shaft and cams.

To remedy this, retard the spark. If this does not remedy the cause an examination of the cylinder should follow, then the bearings. Inspect thoroughly the engine to see if there are any loose parts as indicated, to find any broken valve stems or broken pumps. Inspect also for wear. Tighten all loose parts. Take broken or worn parts to machinist.

Irregular Knock.—An irregular knock is different from a regular and has an entirely different cause. It is usually caused by loose electrical connec-

tions, loose piping, pre-ignition caused by same variable faults, or defective commutator or timer.

Make a thorough inspection of electrical connections for loose piping. Look under discussion for pre-ignition for remedies to prevent this. Inspect commutator or timer for poor adjustment.

Speed Variation—Speed may vary due to misfiring, water in the fuel, irregular supply of gas, magneto slipping, defective fuel pump, defective governor, wear on valve gear, defective times, loose electrical connections, and a poor mixture.

To remedy this read the paragraph on misfiring. Inspect engine for that cause. Empty the fuel tank and put in a fresh supply of clean fuel which is sure not to have water. Note the supply of gas, see if it is regular. See if the magneto belt is slipping. Inspect the fuel pump and governor, wear on valve gears, and make a thorough inspection of the timer. See if some electrical connection is not loose, which would part of the time be in contact and part of the time away. Enrich the mixture.

The above discussion on gas engine troubles is quite complete and if properly followed will give a clue to most any trouble that has any chance whatever to happen. In cases where these do not cover, the engine should be taken to an engine works for treatment.

Miscellaneous Upkeep of a Gas Engine.

There are cértain operations in the care of an engine which have been referred to under “Gas Engine Troubles” but need a little further explanation. These will include the repair of worn parts, repair of cracked or broken parts, repair of warped parts, use of lubricating oil, general care of the engine, and starting and stopping the engine.

Worn Parts.—The principal wearing parts of an engine are the piston rings, the valves and the bearings. If the piston rings are worn they may be replaced with new ones. Disconnect the connecting rods at the crank end and remove the piston by drawing out of the cylinder from below. Remove the old piston rings by springing them open with a screw-driver, put on the new ones and insert the piston, one ring at a time until all the rings are in. Sometimes the piston becomes badly worn. In this case it must be replaced entirely.

When valves are worn and leaking they can be repaired by grinding. If the valve does not show a bright bearing all the way around, it is in need of grinding. On the top of the valve will usually be found a slot which will receive a screw-driver. To grind the valve proceed as follows: oil both seat of palate and seat of the valve. Sprinkle in some floured emery. Then with brace and screw-driven turn the valve backward and forward, occasionally lifting and supplying a new charge of oil and emery. When the valve becomes smooth and bright and turns without apparant grating friction, wipe it clean, examine it and proceed to grind more if necessary. When the grinding is done, wipe all the parts clean and replace the valve. If the spring is toa

strong to allow placing the valve by the hand, compress the spring in a vise and tie it down with wires. When the valve is then put in position, cut the wires, replace the locknut and the valve is ready.

Cracked Parts.—Sometimes the parts of the casings or even the moving parts are cracked from various causes. The water jacket suffers this trouble after freezing. Cracking of the water-jacket usually occurs after freezing and does not interfere in any way with the inside bore of the cylinder. A farmer with the proper tools will be able to make a satisfactory repairing of such an accident if it is not too large. Put a piece of sheet iron plate over the crack extending about an inch on each side. Bore a hole through the sheet into the jacket and tap the holes to receive screws. Clean the crack out with chisel so as to make a V-shaped groove. Saturate with a cream made of white lead and with an asbestos strip to form a mat in the groove. Cover it with the sheet metal and screw down tightly. This must be dried for several days before using. This will make a good temporary patch and will last well under ordinary service.

Parts that are badly broken can not be replaced economically except from the factory or by some means of electric welding or acetylene welding.

When engine is overheated or bearings are overheated, the expansion which takes place is oftentimes so great and uneven that the parts are twisted out of shape or warped. The only way this can be corrected is by turning the machine over to a machinist.

Always use the best lubricating oils. Buy oil of a responsible dealer and take his advice where to use it.

Care of Engine.—There is only one way to make an engine give a life long service and that is to take care of it. One of the first requisites is to keep it clean. Not only should the outside be kept clean, but all bearings and rubbing parts, whether oiled or not, should be carefully inspected from time to time and kept perfectly clean, and nothing but pure clean oil used. All these parts should be thoroughly looked over by the owner every two weeks. Give the batteries and connections the very best of attention. When the engine is not in use, they should always be disconnected or they will run down and be of no service whatsoever.

To Start an Engine-—To a new owner, a little advice in regard to the routine of starting an engine will not come amiss. First, see that the engine is clean and all nuts are tight. Second, examine the fuel supply. Third, see that every part is oiled and adjusted. Fourth, test the battery switch and then close it. Fifth, turn the engine over slowly and note if the valves and piston operate freely. Sixth, admit cooling water from the radiator. Seventh, ctank the engine two or three sharp turns with the ignition closed and the engine will start.

To Stop an Engine.—Shut off either the gas or the spark. The more economical is to shut off the gas. After stopping the engine, close all oil cups, needle valves, and the water cocks which drain the water from the cylinder and from the cooling tank.